

Gurpreet (Gup) Singh

Berkeley, CA | (818) 318-0083 | gupsingh@berkeley.edu | [LinkedIn](#) | [Google Scholar](#)

Education

University of California, Berkeley

December 2022

Bachelor of Arts, Physics

Experience

Research Data Analyst I / Postgraduate Researcher, UC Berkeley Feb 2026-Present; Feb 2023-Aug 2024

- Implemented and benchmarked HMC and microcanonical Langevin Monte Carlo samplers for high-dimensional experimental likelihoods; evaluated convergence, effective sample size, runtime, and posterior agreement on real experimental data
- Developed real-time signal processing and data acquisition software on AMD RFSoc hardware, achieving ~100 dB signal-to-noise ratio for low-noise experimental inference pipelines

Researcher, Leinweber Institute for Theoretical Physics

May 2023-Present

- Built a Bayesian inference framework that combines theoretical model evolution with experimental constraints to estimate parameters across high-dimensional physics models
- Developed computational modeling and numerical methods that led to two published *Physical Review D* papers on chiral gauge theory dynamics

DOE SULI Internship, Lawrence Berkeley National Laboratory

Aug 2023-May 2024

- Processed 100M+ QCD simulation events using NERSC supercomputing GPU clusters, training classifiers with custom loss functions on large-scale datasets for particle physics anomaly detection
- Implemented and benchmarked the weakly supervised method against leading baselines, improving sensitivity by 10× on the reported benchmark; work resulted in a *Physical Review Letters* publication and an invited ML4Jets talk at Caltech.

Research Assistant, UC Berkeley

Aug 2023-Mar 2024

- Built an end-to-end data processing pipeline over millions of particle collision events; deployed ML classifiers that succeeded where traditional methods failed

Projects

Bayesian Monte Carlo Sampler Framework

- Implemented Hamiltonian Monte Carlo with a tuning mass matrix from scratch and a framework to validate posterior agreement, convergence, effective sample size, and runtime using statistical measures

Gitlet

- Built a local version of Git from scratch in Java using custom data structures, supporting core commands (init, add, commit, branch, checkout, rm)

Publications

Incorporating Physical Priors into Weakly Supervised Anomaly Detection – **Physical Review Letters** (ML/statistics)

Generator Based Inference – **Data Science in Science** (ML/statistics)

Exact Results in Chiral Gauge Theories with Flavor – **Physical Review D** (Theoretical Physics)

Dynamics of E_6 Chiral Gauge Theories – **Physical Review D** (Theoretical Physics)

Presentations

ML4Jets, California Institute of Technology, Pasadena, CA

Aug 2025

Invited talk on our weakly supervised machine learning method for anomaly detection in particle physics

Skills

Languages: Python (NumPy, SciPy, PyTorch, TensorFlow), Java, C++/C, SQL, Julia, LaTeX, React, MATLAB

Compute & Tooling: NERSC GPU clusters, SLURM job orchestration, Git, LabVIEW

Hardware: Electrical Circuits, RF Signal Processing, Soldering, G-code

Honors & Awards

Berkeley Transfer Scholarship | The Colorado Space Grant Consortium | Osher Scholarship | Phi Theta Kappa Honors Society